095344

COMPUTER NETWORKS

Course Outcome

CO1: Understand the concept of OSI and TCP/IP reference model
CO2: Understand the basics of data transmission at physical layer.
CO3: Understand the channel allocation using ALOHA, CSMA and CSMA/CD.
CO4: Apply error detection and correction technique to eliminate transmission error.
CO5: Analyze the fixed and variable length address (IPv4) subneting for the given scenarios.
CO6: Understand the design issues of the transport layer.
CO7: Understand the mechanism of protocols at application layer such as FTP, HTTP, Telnet, DNS.
CO8: Understand IPv6 addressing and differentiate it from IPv4.

Printed Pages:4 **University Roll No. .....**

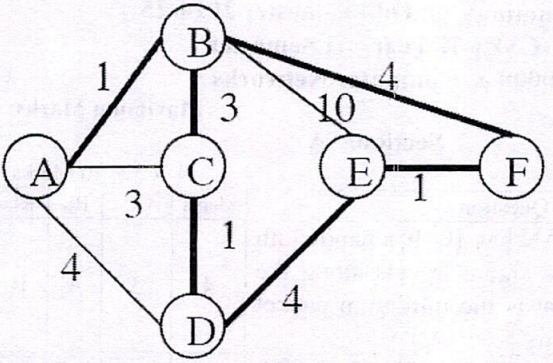
End Term Examination, Odd Semester 2024-25
B. Tech (CSE), II Year, III Semester
BCSC0008 & Computer Networks

Time: 3 Hours **Maximum Marks: 50**

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	A 2 km long broadcast LAN has 10^7 bps bandwidth and uses CSMA/CD. The signal travels along the wire at 2×10^8 m/s. What is the minimum packet size that can be used on this network?	4	3	A	P
2	Consider a LAN with four nodes S1, S2, S3 and S4. Time is divided into fixed-size slots, and a node can begin its transmission only at the beginning of a slot. A collision is said to have occurred if more than one node transmits in the same slot. The probabilities of generation of a frame in a time slot by S1, S2, S3 and S4 are 0.1, 0.2, 0.3 and 0.4, respectively. Determine the probability of sending a frame in the first slot without any collision by any of these four stations.	4	2	E	C
3	The company is also considering the use of satellite communication for connecting its rural offices. However, there is concern about the cost and complexity of installing satellite systems. A. How does satellite communication compare to fiber optic or microwave communication in terms of range, bandwidth, and installation costs?	4	3	A	P

	B. What are the advantages and disadvantages of unguided media like satellite communication for this global network?				
4	In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total length is 400 and the fragment offset value is 300. The position of the datagram, the sequence numbers of the first and the last bytes of the payload, respectively are?	4	5	A	P
5	Consider the network shown below.  A. Show the operation of Dijkstra's (Link State) algorithm for computing the least cost path from F (the rightmost node in the figure below) to all destinations. Also explicitly list all the shortest path routes from F to all destinations that are the result of the algorithm's computation. B. Show the distance table that would be computed by the distance vector algorithm in B.	4	6	An	M

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	A company's email system relies on SMTP to send outbound emails. Employees are able to receive emails but are having trouble sending them. After checking the mail server logs, the administrator finds that the emails are not being relayed to the mail server.	3	7	A	P

	A. What is the role of SMTP in email communication, and why is it essential for sending emails? B. What troubleshooting steps would you follow to identify and fix the issue with email sending? Consider common configuration errors related to SMTP				
7	Describe the role of application layer and session layer of OSI model in detail.	3	7	R	F
8	What is the purpose of the Domain Name System? Discuss the three main divisions of the domain name space.	3	7	U	F
9	In a RSA cryptosystem, a participant A uses two prime numbers p=13 and q=11 to generate his public and private keys. If the public key of A is 37, then the private key of A is? If the plaintext is 5 then show the process of cipher text and plain text generation using RSA cryptosystem.	3	8	A	C
10	Your router has the following IP address on Ethernet0: 172.16.2.1/23. Calculate the Network ID and valid host IDs on the LAN interface attached to the router?	3	5	E	P

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 4 MSS and the threshold at the start of the first transmission is 16 MSS. Assume that a timeout occurs during the fourth and seventh transmission. Find the congestion window size at the end of the eleventh transmission.	5	6	A	P
12	The following is the dump of a TCP header in hexa decimal format: 05320017 00000001 00000000 500207FF 00000000 (i) What is the sequence number?	5	6	An	C

	(ii) What is the destination port number? (iii) What is the acknowledgment number? (iv) What is the window size?				
13	An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: 245.248.128.0/21. The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Calculate the valid IP addresses for the allocation to Organization A and Organization B?	5	8	An	M